

Automatic Segmentation and Landmark Placement for 3D Morphometric Analysis of the Mouse Brain

Beyza Zayim^{1,†}, Emilia Skutunova^{2,3,†}, Taiabur Rahman^{3,†}, Nida Yardim³, Salma Zarfaoui³, Hanzala Daud³, Alienor Vaudene³, Binnaz Yalcin^{2,3*}, Alain Lalande^{1,4*}, Fabrice Meriaudeau^{1*}, and Stephan Collins^{2,3*}

¹ Université Bourgogne Europe, CNRS, ICMUB UMR 6302 Laboratory, Dijon 21000, France

² Institut NeuroMyoGène, INMG-PGNM-CNRS UMR 526, Lyon, France

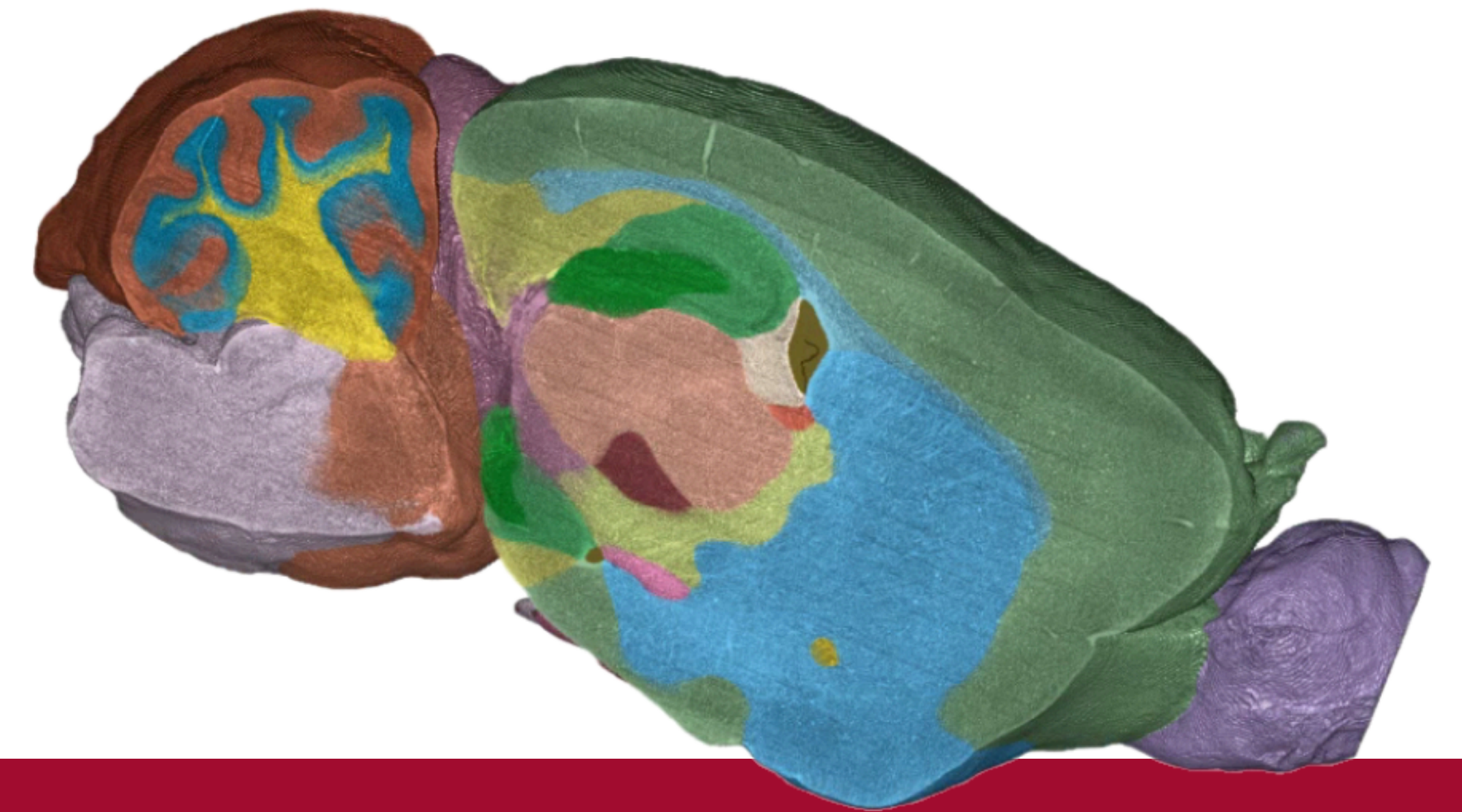
³ University of Bourgogne Europe, Inserm UMR1231, Center of Translational Molecular Medicine, Mouse NeuroGenetics Laboratory, Dijon, France

⁴ Department of Medical Imaging, University Hospital of Dijon, Dijon, France

†,* Equal contribution

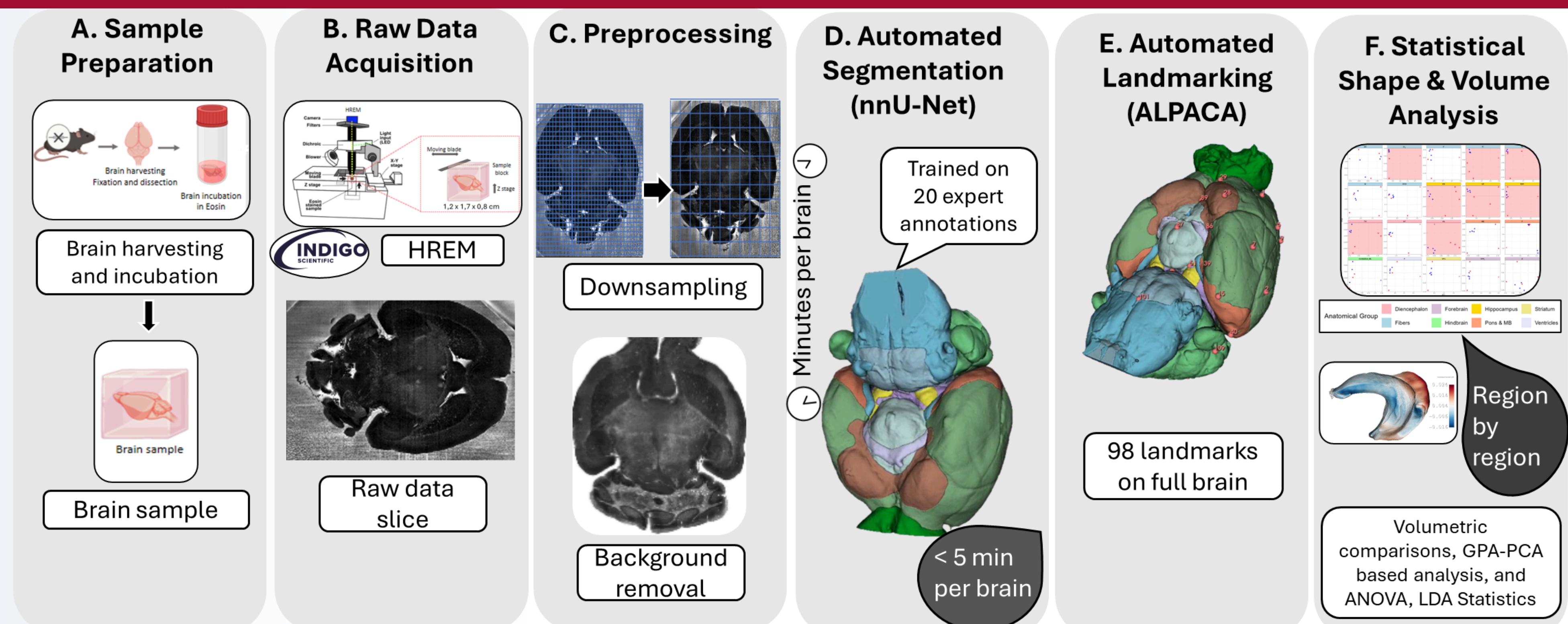
Introduction

- The precise neuroanatomical phenotyping is essential for understanding the genetic basis of neurodevelopmental disorders.
- The project presents an end-to-end pipeline that converts 3D raw HREM data into automatically segmented 27 mice brain regions by using nn-UNet and detect anatomically meaningful landmarks by using ALPACA registration for volumetric and morphometric analysis.



Methods

- nnU-Net default method for the segmentation by using 20 annotated half brains, tested on 38 mice HREM images (Dataset 1: n=8; 4 WT, 4 HOM. Dataset 2: n=30; 13 WT, 13 HET, 4 HOM)
- ALPACA registration method to place 98 landmarks around the segments of the brain.
- Generalized Procrustes Analysis-Principal Component Analysis (GPA-PCA) method in 3D Slicer software by using landmarks, for the shape analysis of the segments of the test data



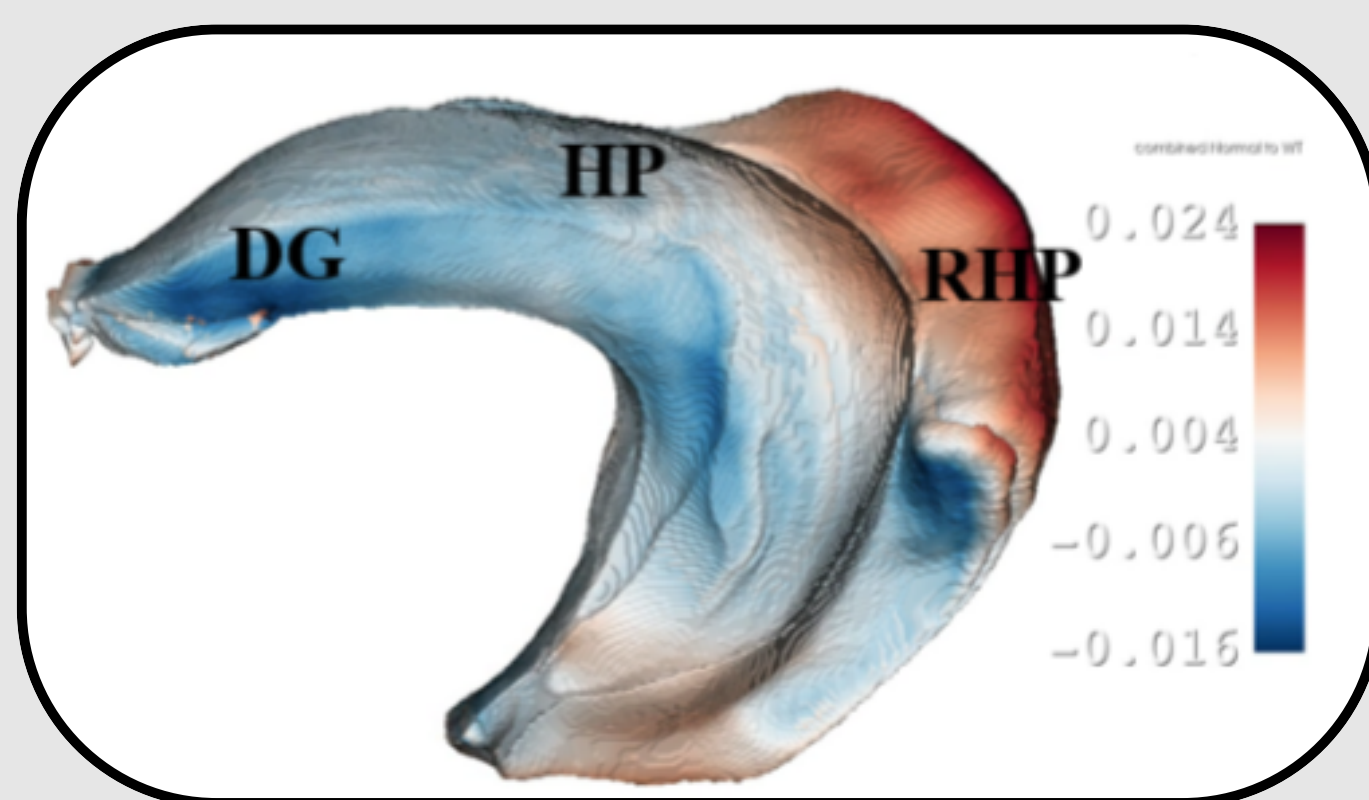
Results

Dataset 1:

- Volumetric**: Significant volume reductions in the **hippocampal formation** and the fimbria (fi) for the homozygous mutant (t-test)
- Shape**: **Hippocampal dysmorphology** in mutant samples compared to controls (ANOVA)

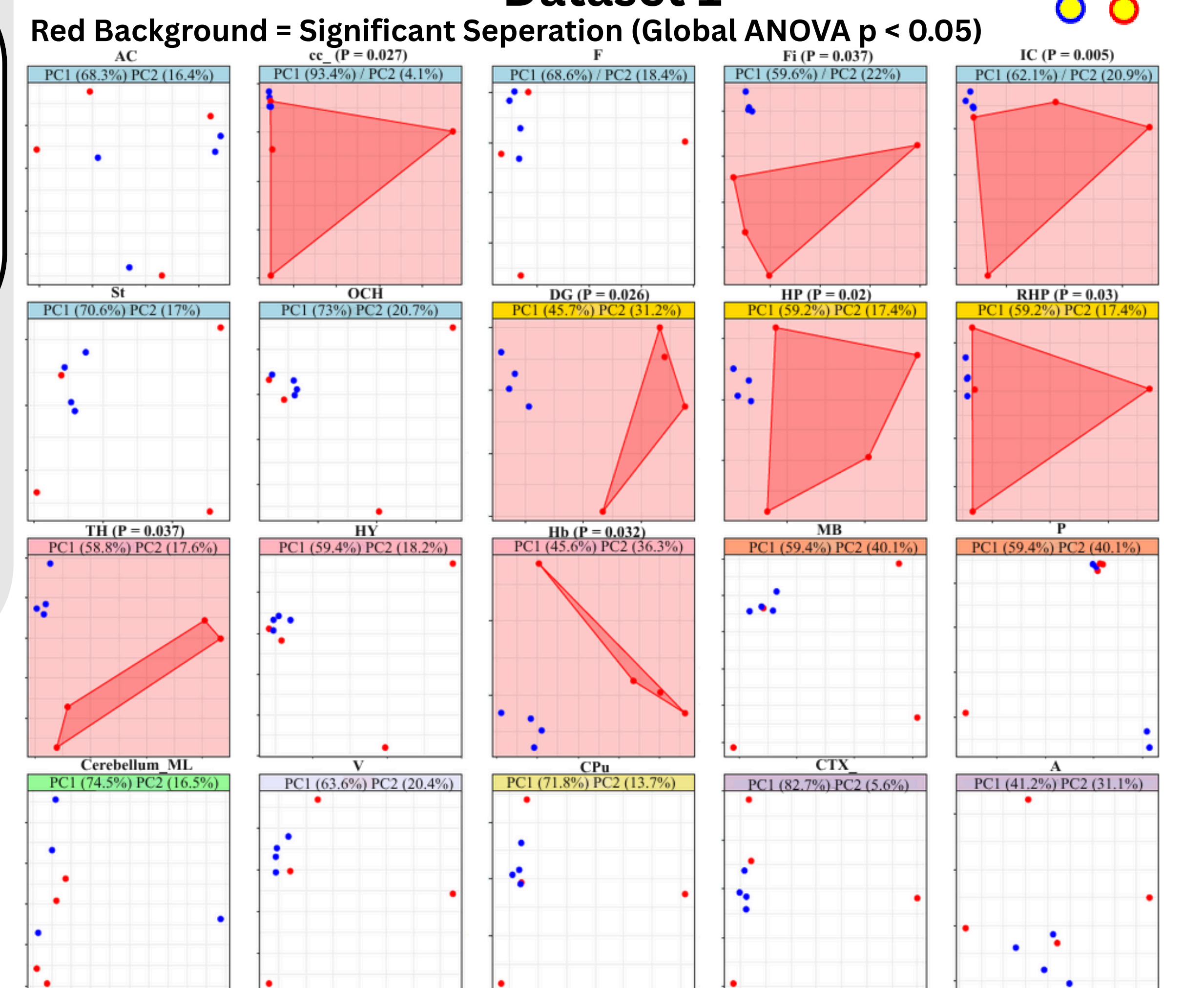
Dataset 2:

- Volumetric**: Smaller sized corpus callosum (cc+), **internal capsule (ic)**, and stria terminalis (st) for homozygous mutant (ANOVA)
- Shape**: **Internal capsule (ic)**, hypothalamus (HY), and ventricles (V) as the regions with the strongest discriminatory power (LDA permutation p < 0.05)

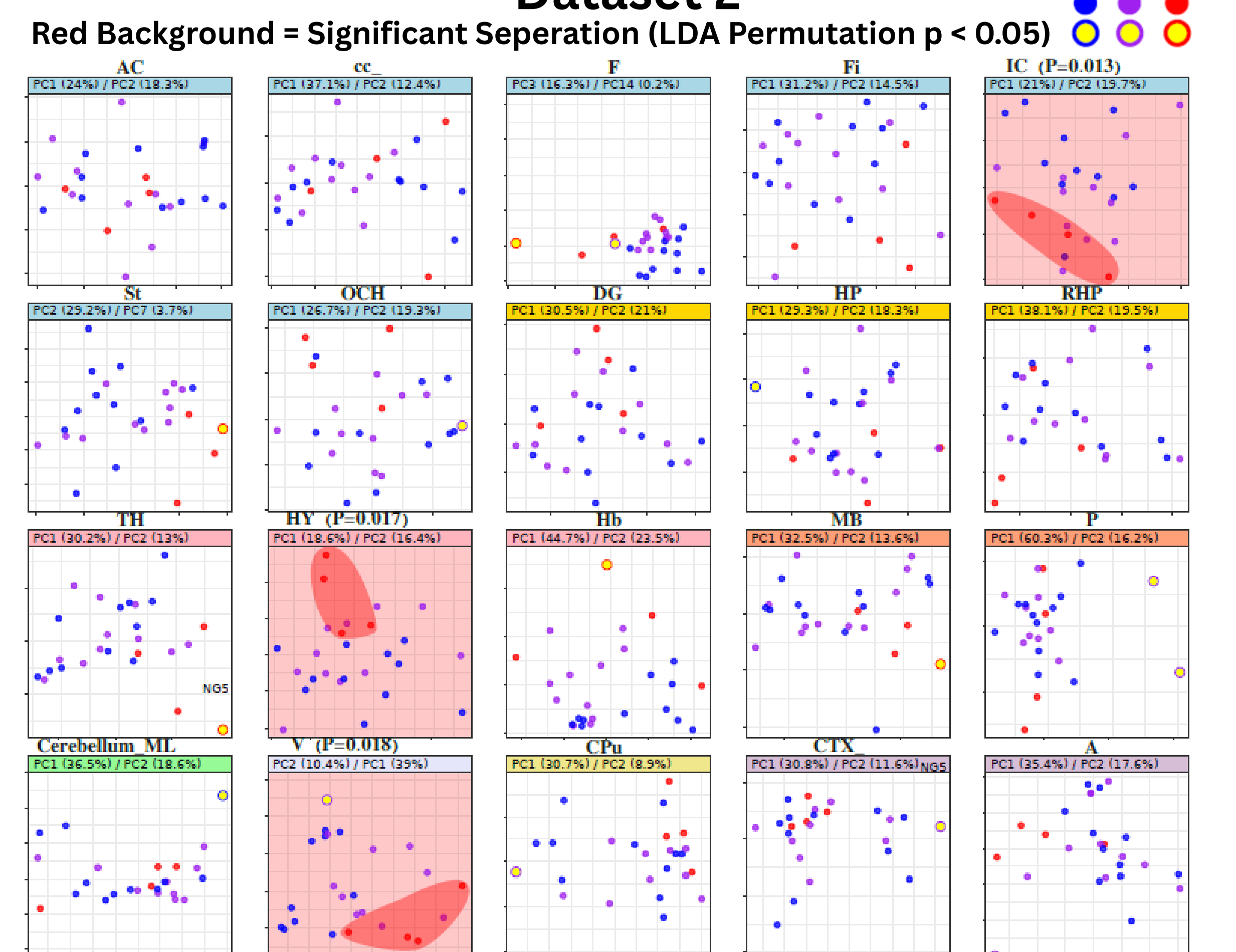


Shape Analysis

Dataset 1



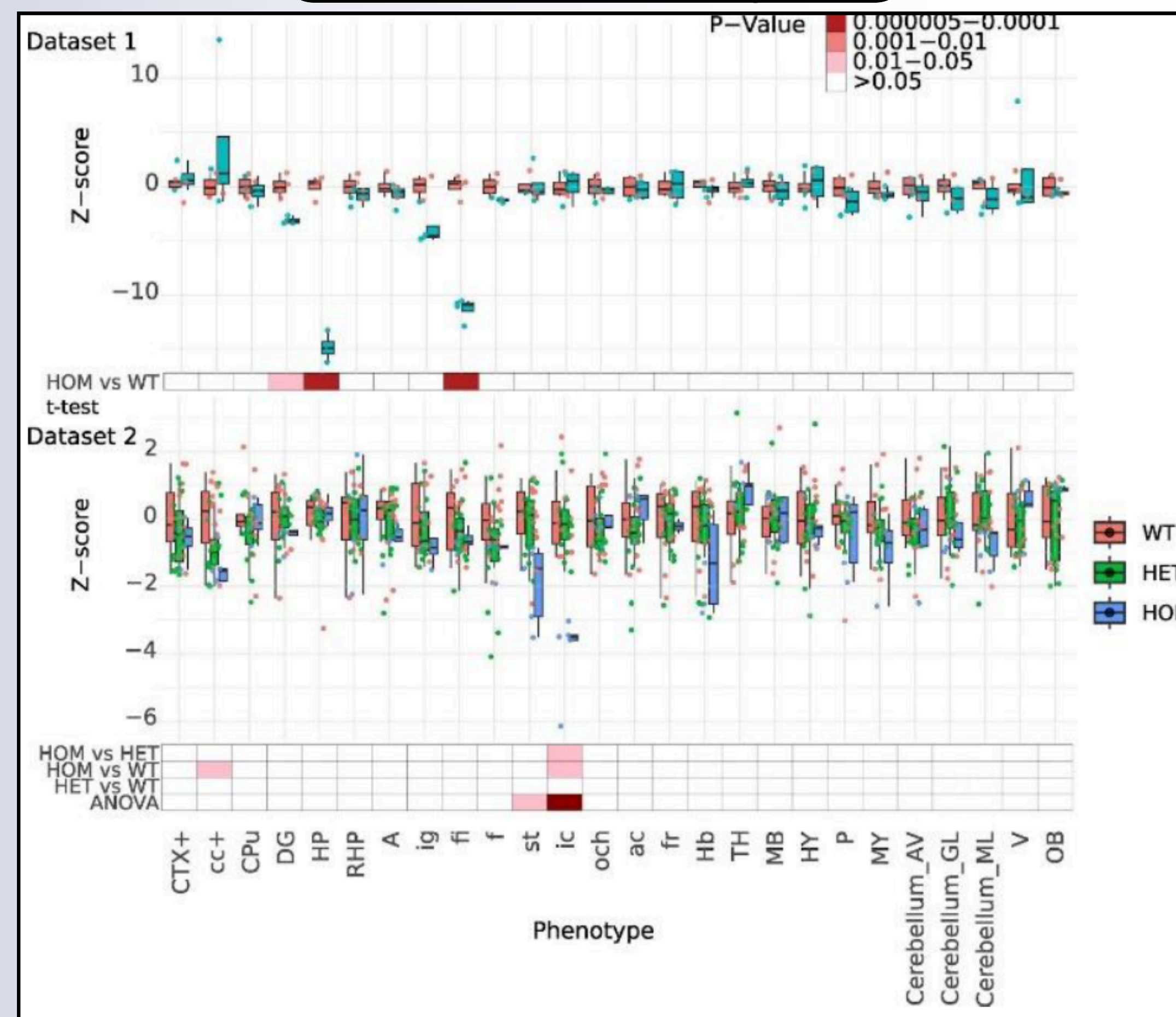
Dataset 2



nnU-Net Validation Results

Region	Dice	HD95
A	0.91 ± 0.02	3.84 ± 1.62
ac	0.89 ± 0.02	3.55 ± 1.97
Cerebellum_AV	0.92 ± 0.02	4.62 ± 0.78
Cerebellum_GL	0.92 ± 0.02	5.23 ± 8.34
Cerebellum_ML	0.96 ± 0.01	6.03 ± 8.22
cc+	0.91 ± 0.02	8.09 ± 2.41
CPu	0.97 ± 0.01	5.73 ± 1.60
CTX+	0.98 ± 0.01	17.86 ± 26.78
DG	0.93 ± 0.02	3.13 ± 0.99
f	0.82 ± 0.08	10.60 ± 27.56
fi	0.92 ± 0.02	4.94 ± 2.05
fr	0.79 ± 0.18	5.15 ± 1.10
Hb	0.89 ± 0.04	4.63 ± 7.67
HP	0.93 ± 0.03	2.79 ± 0.80
HY	0.94 ± 0.02	9.36 ± 22.15
ic	0.88 ± 0.01	5.90 ± 8.73
IG	0.44 ± 0.32	4.91 ± 1.09
MB	0.97 ± 0.00	5.19 ± 0.91
MY	0.96 ± 0.01	5.54 ± 2.39
OB	0.91 ± 0.01	8.23 ± 3.29
och	0.88 ± 0.07	6.12 ± 4.54
P	0.95 ± 0.02	3.01 ± 1.06
RHP	0.92 ± 0.02	1.79 ± 0.46
st	0.84 ± 0.05	1.53 ± 0.31
TH	0.96 ± 0.02	8.93 ± 15.72
V	0.93 ± 0.04	4.63 ± 4.77
border	0.96 ± 0.01	1.21 ± 0.29

Volumetric Analysis



Conclusion

- Systematic end-to-end 3D phenotyping for large-scale genetic studies of mouse brain morphology is provided with this study.
- Automatic segmentation of the brain regions is highly time-efficient.